

EXAM 7 – SPRING 2011

12. (3.25 points)

Given the following information to be used in the valuation of a firm:

Firm's ROE:	15%
Firm's plowback ratio:	90%
Market return:	9%
Yield on long-term T-bonds	5.2%
Liquidity premium for long-term T-bonds:	1.2%
Average growth rate for firms in the same industry:	4.2%
Average Beta for firms in the same industry:	1.4
Beta for this firm:	2.0

- The above values are assumed to continue indefinitely into the future.

Projections for the firm's net income after tax:

<u>Calendar Year</u>	<u>Net Income after Tax (\$000)</u>
2011	\$10,100
2012	11,464
2013	13,011

a. (0.75 point)

Demonstrate why it is reasonable for this firm's Beta to be significantly higher than the average Beta for firms in its industry.

b. (2 points)

Use the dividend discount model to estimate the value of this firm as of December 31, 2010.

c. (0.5 point)

Discuss how a company's growth rate may affect its dividend payout rate and risk-adjusted discount rate.

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Sample Solutions
Exam 07, 2011, Question 12

Solution 1

a)

Firm's Dividend Growth Rate = $g = \text{Plowback Ratio} \times \text{ROE} = 90\% \times 15\% = 13.5\%$
 The average industry growth rate is given as 4.2%

Since the firm's growth rate is significantly higher than the industry's you can surmise that this firm is riskier than the average firm in the industry and therefore should have a Beta greater than the industry average.

b)

Determine the discount rate to use (CAPM):

Risk Free Rate = Yield on T-Bonds - Liquidity Premium on T-Bonds = $5.2\% - 1.2\% = 4.0\%$

Discount Rate = $k = (\text{Risk Free Rate}) + \text{Beta} \times (\text{Equity Risk Premium-diff. between expected market return and risk free})$
 $= 4.0\% + 2.0 \times (9.0\% - 4.0\%) = 14.0\%$

Determine the dividends and terminal value, then discount to get the company valuation:

Div = Net Income $\times$ Plowback ratio

	$10,100 \times (1-90\%)$	
DIV2011:	=	1,010
DIV2012:	$11,464 \times 10\% =$	1,146
DIV2013:	$13,011 \times 10\% =$	1,301

$\text{Terminal Value} = \frac{E(\text{Dividend}_{2014}) \times (1+g)}{k-g} = \frac{(1301) \times (1+13.5\%)}{14.0\% - 13.5\%} = 295,377$

Value of Firm as of 12/31/10:

= $\text{Exp}(\text{Dividend 2011})/(1+k) + \text{Exp}(\text{Dividend 2012})/(1+k^2) + \text{Exp}(\text{dividend 2013})/(1+k^3)$
 + Terminal Value/($1+k^3$)
 $= 1,010/(1.14) + 1,146/(1.14^2) + 1,301/(1.14^3) + 295,377/(1.14^3)$
 $= 202,017$

c)

High dividend payout rates are unlikely to be sustainable and so the effects of high growth rates are likely to be offset by lower dividend amounts. The dividend payments for firms with high growth rates are likely to be riskier than those of firms with low growth rates. As a result, the effect of high growth rates are likely to be offset by discounting dividends to present value using higher risk adjusted discount rates.